

# PROMPT FOR RESOLVING THE LONELY RUNNER CONJECTURE

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ABSTRACT. This document contains a full frontier-model research prompt for resolving the Lonely Runner Conjecture for every number of runners. The stationary-runner formulation has been verified through 12 nonzero relative speeds, corresponding to 13 total runners; the task is a uniform proof or an exact counterexample. Status snapshot: 20 July 2026.

## 1. Prompt

### Current task statement

For  $x \in \mathbb{R}$ , let  $\|x\|$  denote the distance from  $x$  to the nearest integer. In the stationary-runner formulation, the Lonely Runner Conjecture states:

For every  $k \geq 1$  and all distinct nonzero integers  $u_1, \dots, u_k$ , there exists  $t \in \mathbb{R}$  such that  $\|tu_i\| \geq 1/(k+1)$  for all  $i$ .

This is equivalent to the usual statement for  $k+1$  runners with distinct constant speeds after subtracting the speed of the runner being observed.

Resolve the conjecture completely for all  $k$ . As of the status date for this prompt, the theorem is known through  $k=12$ , corresponding to 13 total runners. Do not confuse the conjecture with shifted, view-obstruction, or lonely-vector variants that have different truth values.

### Complete-resolution standard

A complete resolution must give exactly one of the following:

- Affirmative alternative: a proof valid for every integer  $k \geq 1$  and every admissible speed tuple;
- Negative alternative: a specific finite  $k$  and exact speed tuple for which no real time satisfies all inequalities, together with a rigorous universal-in- $t$  nonexistence proof;
- in either direction, explicitly reconcile the stationary-runner and total-runner conventions, all normalizations, and equality at the boundary  $1/(k+1)$ .

### What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- verification for any fixed additional number of runners without a uniform theorem for all  $k$ ;
- a probabilistic or numerical search over time that misses narrow feasible intervals;
- a proof for almost all speed tuples, generic tuples, bounded speeds, or a special divisibility class;
- a result for the shifted Lonely Runner Conjecture or another inequivalent variant;
- an induction whose step assumes a simultaneous approximation statement as strong as the original conjecture;
- a reduction to a finite computation without a proved bound uniform in arbitrary input speeds and arbitrary  $k$ .

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

#### Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- torus dynamics and covering: view  $t(u_1, \dots, u_k) \bmod 1$  as a one-parameter subgroup avoiding the union of forbidden coordinate slabs;
- simultaneous Diophantine approximation, geometry of numbers, transference, lattice coverings, and zonotope or polytope formulations;
- Fourier analysis, trigonometric polynomials, Riesz products, positive-definite functions, and exact integration identities;
- additive combinatorics of speed sets, subset sums, modular concentration, gcd structure, and density increment arguments;
- finite reduction through gap patterns, view-obstruction geometry, inclusion-exclusion, and certified cell decompositions;
- extremal and minimal-counterexample analysis under scaling, sign changes, gcd normalization, deletion, and speed compression;
- algorithmic experiments that discover exact threshold lemmas or finite forbidden patterns, followed by independent proof;
- induction on  $k$  using a genuinely stronger invariant or slack parameter rather than a circular restatement of the target inequality.

#### Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- check the convention carefully:  $k$  nonzero relative speeds correspond to  $k+1$  total runners and threshold  $1/(k+1)$ ;
- prove all reductions preserve distinctness, nonzero speeds, and the simultaneous nature of the inequalities;
- handle negative speeds, common gcd, permutation, time periodicity, and boundary equality exactly;
- for a proposed counterexample, cover the entire time circle with exact forbidden intervals and verify

the cover symbolically;

- distinguish "there exists a time for one chosen runner from "every runner is lonely at some possibly different time and prove the equivalence used;
- audit any finite-search bound for dependence on  $k$  and on the magnitudes of the speeds;
- test general lemmas against exhaustive small- $k$  data and actively search for counterexamples to every strengthening introduced.

#### Required final artifact

The final response must include a conventional proof and all exact supporting artifacts:

- for an affirmative solution, a complete uniform argument with every auxiliary lemma stated quantitatively;
- for a negative solution, the exact speed tuple and a symbolic certificate that no time works;
- any computer-assisted finite decomposition, interval cover, or optimization certificate in a machine-checkable format;
- source code, exact rational data, hashes, and independent checker instructions;
- a short section translating the result between the stationary and moving-runner formulations.

Public search may be used for standard formulations, known small- $k$  results, and ordinary Diophantine-approximation tools. Do not search for a purported final resolution or answer only that verification currently reaches 13 total runners.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.